3. It is also used as fire extinguisher and as a propellent gas.
4. It is used in the production of carbonated beverages and in the production of foam.

2.3.7 Silicon tetrachloride:

Preparation:

Silicon tetrachloride can be prepared by passing dry chlorine over an intimate mixture of silica and carbon by heating to 1675 K in a porcelain tube

$$SiO_2 + 2C + 2Cl_2 \longrightarrow SiCl_4 + 2CO$$

On commercial scale, reaction of silicon with hydrogen chloride gas occurs above 600 K

$$Si + 4HCl \longrightarrow SiCl_4 + 2H_2$$

Properties:

Silicon tetrachloride is a colourless fuming liquid and it freezes at -70 ^{0}C

In moist air, silicon tetrachloride is hydrolysed with water to give silica and hydrochloric acid.

$$SiCl_4 + 4H_2O \longrightarrow 4HCl + Si(OH)_4$$

When silicon tetrachloride is hydrolysed with moist ether, linear perchloro siloxanes are formed $[Cl\text{-}(Si\,Cl_2O)_nSiCl_3$ where n=1-6.

Alcoholysis

The chloride ion in silicon tetrachloride can be substituted by nucleophile such as OH, OR, etc.. using suitable reagents. For example, it forms silicic esters with alcohols.

$$SiCl_4 + C_2H_5OH \longrightarrow \underset{\text{Tetraethoxy silane}}{Si(OC_2H_5)_4} + 2Cl_2$$

Ammonialysis.

Similarly silicon tetrachloride undergoes ammonialysis to form chlorosilazanes.

$$2SiCl_4 + NH_3 \xrightarrow[\text{Ether}]{330\ K} Cl_3Si\text{-}NH\text{-}SiCl_3$$

Uses:

1. Silicon tetrachloride is used in the production of semiconducting silicon.
2. It is used as a starting material in the synthesis of silica gel, silicic esters, a binder for ceramic materials.

2.3.8 Silcones:

Silicones or poly siloxanes are organo silicon polymers with general empirical formula (R_2SiO). Since their empirical formula is similar to that of ketone (R_2CO), they were named "silicones". These silicones may be linear or cross linked. Because of their very high thermal stability they are called high –temperature polymers.

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